THE GYSIN PULLBACK OF THE NORMAL CONE OF Z IN X IS THE NORMAL CONE OF Z IN Y

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FEB (sources/stacks-chow/pinned-original-source.tex line 12260).

Allowed standard prerequisites (editorial).

- The blowing up of infinity(Z) and the rational map induced by the inclusion of ideals. Same chapter, Section labelled section-blowup-Z-first, invoked at proof lines 12268-12270; Constructions, Lemma labelled constructions-lemma-morphism-relative-proj, cited at proof lines 12276-12278. Produces the open U in W mapping to W'.
- Strict transform of P^1_Y inside W and its special fibre. Divisors, Lemma labelled divisors-lemma-strict-transform, cited at proof lines 12289-12292. Identifies W" with the blowup of P^1_Y in infinity(Z) and (W" cap U)_infinity with C_Z Y.
- Gysin maps of effective Cartier divisors and their commutation. Same chapter, Lemma labelled lemma-gysin-bivariant, cited at proof lines 12295-12296, and Lemma labelled lemma-gysin-commutes-gysin, cited at proof lines 12317-12319. Supplies the bivariant classes i^*, (i'_infinity)^* and i_infinity^* and lets their order be exchanged.
- Vanishing at infinity and the factorization of the fibre of i over infinity. Same chapter, Remark labelled remark-restriction-bivariant, cited at proof lines 12298-12300, and the notation of the section at proof lines 12319-12321. Gives i_infinity^* i^*[U] = [C_Z Y]_n and the description of the fibre through o.

Source reference. Original section 0FBG: sources/stacks-chow/pinned-original-source.tex, source lines 11646–11646. Same-chapter pointer into the author environment lemma at source line 11646; invoked at proof line 12270 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original lemma 07ZG: sources/stacks-chow/tags.txt, source lines 10146–10146. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12279 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 080E: sources/stacks-chow/tags.txt, source lines 10179–10179. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 12290 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0FE9: sources/stacks-chow/pinned-original-source.tex, source lines 11678–11678. Same-chapter pointer into the author environment lemma at source line 11678; invoked at proof line 12295 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B79: sources/stacks-chow/pinned-original-source.tex, source lines 6149–6149. Same-chapter pointer into the author environment lemma at source line 6149; invoked at proof line 12296 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0F9Z: sources/stacks-chow/pinned-original-source.tex, source lines 6184–6184. Same-chapter pointer into the author environment remark at source line 6184; invoked at proof line 12299 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B73: sources/stacks-chow/pinned-original-source.tex, source lines 5496–5496. Same-chapter pointer into the author environment lemma at source line 5496; invoked at proof line 12318 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *With notation as above we have*

$$o^*[C_Z X]_n = [C_Z Y]_{n-1}$$

in $\mathrm{CH}_{n-1}(Y \times_{o,C_Y X} C_Z X)$.

Proof. Denote $W \to \mathbf{P}^1_X$ the blowing up of $\infty(Z)$ as in Section 0FBG. Similarly, denote $W' \to \mathbf{P}^1_X$ the blowing up of $\infty(Y)$. Since $\infty(Z) \subset \infty(Y)$ we get an opposite inclusion of ideal sheaves and hence a map of the graded algebras defining these blowups. This produces a rational morphism from W to W' which in fact has a canonical representative

$$W \supset U \longrightarrow W'$$

See Constructions, Lemma 07ZG. A local calculation (omitted) shows that U contains at least all points of W not lying over ∞ and the open subscheme $C_Z X$ of the special fibre. After shrinking U we may assume $U_\infty = C_Z X$ and $\mathbf{A}^1_X \subset U$. Another local calculation (omitted) shows that the morphism $U_\infty \to W'_\infty$ induces the canonical morphism $C_Z X \to C_Y X \subset W'_\infty$ of normal cones induced by the inclusion of ideals sheaves coming from $Z \subset Y$. Denote $W'' \subset W$ the strict transform of $\mathbf{P}^1_Y \subset \mathbf{P}^1_X$ in W. Then W'' is the blowing up of $\mathbf{P}^1_Y$ in $\infty(Z)$ by Divisors, Lemma 080E and hence $(W'' \cap U)_\infty = C_Z Y$.

Consider the effective Cartier divisor $i : \mathbf{P}^1_Y \to W'$ from (0FE9) and its associated bivariant class $i^* \in A^1(\mathbf{P}^1_Y \to W')$ from Lemma 0B79. We similarly denote $(i'_\infty)^* \in A^1(W'_\infty \to W')$ the Gysin map at infinity. Observe that the restriction of i'_∞

(Remark 0F9Z) to U is the restriction of $i_\infty^* \in A^1(W_\infty \to W)$ to U. On the one hand we have

$$(i'_\infty)^* i^*[U]_{n+1} = i_\infty^* i^*[U]_{n+1} = i_\infty^*[(W'' \cap U)_\infty]_{n+1} = [C_Z Y]_n$$

because i_∞^* kills all classes supported over ∞, because $i^*[U]$ and $[W'']$ agree as cycles over $\mathbf{A}^1$, and because $C_Z Y$ is the fibre of $W'' \cap U$ over ∞. On the other hand, we have

$$(i'_\infty)^* i^*[U]_{n+1} = i^* i_\infty^*[U]_{n+1} = i^*[U_\infty] = o^*[C_Y X]_n$$

because $(i'_\infty)^*$ and i^* commute (Lemma 0B73) and because the fibre of $i : \mathbf{P}^1_Y \to W'$ over ∞ factors as $o : Y \to C_Y X$ and the open immersion $C_Y X \to W'_\infty$. The lemma follows. $\square$